Influence and Isoperimetric Inequalities on Matroids

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Introduction. Isoperimetric inequalities, which relate the area to the length of its boundary, have been widely studied in mathematics. For example, the isoperimetric inequality states that for a closed curve with length L, the area A of the region satisfies $4\pi A \le L^2$ in two-dimensional Euclidean space.

We study influence and isoperimetric inequalities on the Boolean hypercube $\Omega = \{\pm 1\}^n$ with the distribution μ over Ω. For a subset $S \subseteq \Omega$, the analogue of its volume is measured by $\mu(S)$. In the following, we represent a subset $S \subseteq \Omega$ by a Boolean function $f : \Omega \mapsto \{\pm 1\}$ ($f^{-1}(+1) = S$). Boolean functions naturally model voting rules, graph properties, and set families. For example, in a voting rule, each coordinate records an individual vote, $f(x)$ gives the collective decision, and S represents the winning coalitions. In this notation, $\mathrm{Var}_\mu[f] = 4\mu(S)(1-\mu(S)) \ge 2\min\{\mu(S), 1-\mu(S)\}$ serves as the analogue of a volume term, measuring the smaller side of the partition with fixed boundary.

There are several natural notions of boundary on the Boolean hypercube. Two classical ones are based on the *influence* and the *sensitivity* of coordinates. We introduce the influence of the i-th coordinate and the sensitivity with respect to the Boolean function f as follows:

$$\mathsf{Inf}_{\mu,i}(f) = \Pr_{\mu}\left[f(x) \neq f(x^i)\right] \quad \text{and} \quad s_\mu(f)(x) = \sum_{i=1}^{n} \mathbb{1}[f(x) \neq f(x^i)],$$

where x^i is obtained by flipping the sign of the i-th coordinate of x. The influence $\mathsf{Inf}_{\mu,i}(f)$ measures the volume affected by flipping the i-th coordinate while the sensitivity counts how many coordinates change the membership for a given x. In our voting model, they correspond to the importance of the i-th voter and the number of voters who can individually change the outcome of an election, respectively. The *surface area*, $\mathsf{SA}_\mu(f) := \mathbb{E}_\mu\left[\sqrt{s_\mu(f)}\right]$, is the boundary area based on square-root sensitivity. For the uniform measure on the Boolean hypercube, $\mathsf{SA}_\mu(f)$ corresponds to the Houdré–Tetali isoperimetric constant $\varphi_{1/2}$, which interpolates between edge conductance and vertex expansion [LT24]. These quantities have a natural interpretation on the hypercube graph. For a subset of vertices $S = f^{-1}(+1)$, both the *total influence* $\sum_{i=1}^n \mathsf{Inf}_{\mu,i}(f)$ and the surface area $\mathsf{SA}_\mu(f)$ quantify the size of the edge boundary $\partial S = \{(i,x) : f(x) \neq f(x^i)\}$ from different perspectives.

Let us consider μ to be the uniform distribution. One fundamental isoperimetric inequality is the Poincaré inequality: for Boolean function $f : \Omega \mapsto \{\pm 1\}$, $\mathrm{Var}_\mu[f] \le \sum_{i=1}^n \mathsf{Inf}_{\mu,i}(f)$, which guarantees that $\max_{i\in[n]} \mathsf{Inf}_{\mu,i}(f) \ge \mathrm{Var}_\mu[f]/n$. In this proposal, we focus primarily on the following inequalities, for any Boolean function $f : \Omega \mapsto \{\pm 1\}$:

1. **Kahn–Kalai–Linial (KKL) inequality** [KKL88]: $\max_{i\in[n]} \mathsf{Inf}_{\mu,i}(f) \gtrsim \dfrac{\log n}{n} \mathrm{Var}_\mu[f]$.
2. **Talagrand inequality** [Tal93]: $\mathsf{SA}_\mu(f) \gtrsim \mathrm{Var}_\mu[f] \sqrt{\log \dfrac{e}{\mathrm{Var}_\mu[f]}}$.

For a voting rule, KKL guarantees a voter with influence $\Omega((\log n)/n)$, a logarithmic improvement over the Poincaré inequality. These inequalities have a wide range of applications [FK96, GG11, KLMM24, KMS18], such as sharp-threshold phenomena and voting models with correlated preferences.

Previous Work and Proposed Problem. Most previous work establishes influence and isoperimetric inequalities for uniform or biased product distributions [Tal93, KKL88, CL12]. KKL and other influence-type inequalities have also been extended to certain Schreier graphs [OW13] and Markov random fields on bounded-degree graphs under suitable mixing assumptions [KLMM24]. In my recent

work with Zongchen Chen and Xinyuan Zhang [CCZ26], we established these inequalities for distributions with full support that satisfy marginal boundedness and a Dobrushin-type condition [Dob68], which controls dependence among coordinates.

A limitation of our hypercube argument is its reliance on full support. It therefore does not directly apply to distributions with constrained support. An important exception is the uniform distribution over matroid independent sets. We say $\mathcal{M} = (S, \mathcal{I})$ is a matroid if $\mathcal{I} \subseteq 2^S$ is downward-closed and satisfies the exchange property: $\forall A, B \in \mathcal{I}, |A| > |B| \implies \exists u \in A \setminus B, B \cup \{u\} \in \mathcal{I}$. The members of $\mathcal{I}$ are its independent sets. Matroids capture a fundamental combinatorial structure in theoretical computer science. For example, graphic matroids describe acyclic subgraphs. Studying influence and isoperimetric inequalities on matroid independent sets provides insights into Boolean function analysis beyond the Boolean hypercube [Hop26]. In the graphic case, isoperimetric inequalities on graphic matroids would help quantify the stability of graph properties over acyclic subgraphs. This motivates us to explore the influence and isoperimetric inequalities on matroid independent sets. Formally, we would like to answer the following problem.

Proposed Problem. *Do the KKL inequality and Talagrand's inequality hold for the uniform distribution over matroid independent sets?*

More broadly, the proposed problem is a first step toward analogous inequalities for high-dimensional expanders [GK23], determinantal point processes [Tas13], and spin systems with hard constraints [CSV23], such as the hard-core model (vertex-weighted independent sets) and graph colorings. Further targets include Talagrand's L^1–L^2 inequality [Tal94], which strengthens KKL, and the Eldan–Gross inequality [EG22], which incorporates both KKL and surface-area inequalities. Our recent work [CCZ26] also established these stronger inequalities under appropriate assumptions.

Proposed Approach. Markov semigroup methods provide a framework for proving influence and isoperimetric inequalities [CL12, OW13, IZ26]. Our work [CCZ26] utilizes the similar idea together with gradient estimates. We aim to apply the same framework to address the proposed problem.

A *Markov semigroup* $(P_t)_{t\geq 0}$ describes the time evolution of a Markov process and is generated by an operator L ($P_t = e^{tL}$): for any $f : \Omega \mapsto \mathbb{R}$ and $x \in \Omega$, $Lf(x) = \sum_{i=1}^n q_i(x)(f(x^i) - f(x))$. Here $\{q_i\}_{i\in[n]}$ is called the *transition rates* of the Markov semigroup, which is always assumed to be *reversible* with respect to μ: $\mu(x)q_i(x) = \mu(x^i)q_i(x^i)$. Given the Markov semigroup $(P_t)_{t\geq 0}$, the corresponding strong gradient estimate holds with a constant ρ if for any function $f : \Omega \mapsto \mathbb{R}$ and $t \geq 0$,

$$\|\nabla^\star P_t f\|_2 \leq e^{-\rho t} P_t \left(\|\nabla^\star f\|_2\right),$$

where the discrete gradient of f is defined as $\nabla^\star f(x) = [q_i(x)(f(x^i) - f(x))]_{i\in[n]}$. Through the potential function and derivative techniques, the gradient estimate can be equivalently written as a functional inequality, which is known as the reinforced Bakry–Émery condition [BÉ85]. This can be formulated as a local matrix eigenvalue problem [CKLP22]. In [CCZ26], the implication from the gradient estimates to isoperimetric inequalities was established. By analyzing commutators between the generator and discrete gradient, we derived sufficient local matrix criteria for gradient estimates and verified them under suitable weak-dependence assumptions on μ.

In our previous work, we chose transition rates $q_i(x) = \frac{1}{2}\sqrt{\mu(x^i)/\mu(x)}$ satisfying the square identity $q_i(x)q_j(x^i) = q_j(x)q_i(x^j)$ and used the commutativity of coordinate flips $(x^i)^j = (x^j)^i$ to establish the commutator identity $(L\nabla^\star - \nabla^\star L)f = H \cdot \nabla^\star f$, where $H : \Omega \mapsto \mathbb{R}^{n\times n}$. However, for distributions supported on matroid independent sets, coordinate flips may leave the support and the commutativity of elements may not hold, making the square identity fail. Establishing an appropriate commutator identity is therefore a central challenge. One potential approach is to redefine gradients and apply a model-based transition rates. The directed-derivative identities for birth–death dynamics in [GJM+26] and the local Bochner framework in [CL26] suggest a possible route to derive a new commutator identity and establish the gradient estimate.

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